

数学物理方程(理科版)

郑神州

北京交通大学数学系

Email: shzhzheng@bjtu.edu.cn

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第二章 波动方程

§3. 初值问题（高维情形）

1. 齐次三维波动方程Cauchy问题

考虑齐次三维波动方程Cauchy问题:

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \left(\frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} + \frac{\partial^2 u}{\partial x_3^2} \right), & x \in \mathbb{R}^3, t > 0 \\ u(x, 0) = \varphi(x), & x \in \mathbb{R}^3, \\ u_t(x, 0) = \psi(x), & x \in \mathbb{R}^3. \end{cases} \quad (1)$$

3. 一般的三维波动方程的Poisson公式

利用**球平均法**. 设 $M(x_1, x_2, x_3)$ 为心, r 为半径球面 S_r^M 上的平均函数:

$$\begin{aligned}\bar{u}(r, t) &= \frac{1}{4\pi r^2} \iint_{S_r^M} u(\xi, \eta, \zeta, t) dS \\ &= \frac{1}{4\pi} \iint_{S_1^M} u(\xi, \eta, \zeta, t) d\omega,\end{aligned}$$

其中

$$\xi = x_1 + r \sin \theta \cos \varphi$$

$$\eta = x_2 + r \sin \theta \sin \varphi$$

$$\zeta = x_3 + r \cos \theta$$

$$dS = r^2 \sin \theta d\theta d\varphi, \quad d\omega = \sin \theta d\theta d\varphi.$$

对三维波动方程作球平均

设 V_r^M 是 $\mathbb{R}^3$ 中以 M 为心、 r 为半径的任一球体, 其边界球面 S_r^M . 可以验证

$$\bar{u}(r, t) = \frac{1}{4\pi r^2} \iint_{S_r^M} u(\xi, \eta, \zeta, t) dS = \frac{1}{\frac{4}{3}\pi r^3} \iiint_{V_r^M} u(\xi, \eta, \zeta, t) dS.$$

由积分中值定理, 显然

$$u(M, t) = u(x_1, x_2, x_3, t) = \lim_{r \rightarrow 0} \bar{u}(r, t).$$

对齐次三维波动方程在 V_r^M 作积分

$$\iiint_{V_r^M} \frac{\partial^2 u(x'_1, x'_2, x'_3, t)}{\partial t^2} dV = a^2 \iiint_{V_r^M} \left[\frac{\partial^2 u(x'_1, x'_2, x'_3, t)}{\partial x_1'^2} + \frac{\partial^2 u(x'_1, x'_2, x'_3, t)}{\partial x_2'^2} + \frac{\partial^2 u(x'_1, x'_2, x'_3, t)}{\partial x_3'^2} \right] dV$$

$$\begin{aligned}
\text{上述右式} &= a^2 \iiint_{V_r^M} \left[\frac{\partial}{\partial x'_1} \left(\frac{\partial u(x'_1, x'_2, x'_3, t)}{\partial x'_1} \right) + \frac{\partial}{\partial x'_2} \left(\frac{\partial u(x'_1, x'_2, x'_3, t)}{\partial x'_2} \right) + \frac{\partial}{\partial x'_3} \left(\frac{\partial u(x'_1, x'_2, x'_3, t)}{\partial x'_3} \right) \right] dV \\
&= a^2 \iint_{S_r^M} \frac{\partial u(\xi, \eta, \zeta, t)}{\partial n} dS \\
&= a^2 \iint_{S_1^M} \frac{\partial u(\xi, \eta, \zeta, t)}{\partial n} r^2 d\omega = a^2 r^2 \iint_{S_1^M} \frac{\partial u(\xi, \eta, \zeta, t)}{\partial r} d\omega \\
&= 4\pi a^2 r^2 \frac{\partial}{\partial r} \left[\frac{1}{4\pi} \iint_{S_1^M} u(\xi, \eta, \zeta, t) d\omega \right] \\
&= 4\pi a^2 r^2 \frac{\partial \bar{u}(r, t)}{\partial r}.
\end{aligned}$$

$$\begin{aligned}
\text{上式左端} &= \iiint_{V_r^M} \frac{\partial^2 u(x'_1, x'_2, x'_3, t)}{\partial t^2} dV = \frac{\partial^2}{\partial t^2} \iiint_{V_r^M} u(x'_1, x'_2, x'_3, t) dV \\
&= \frac{\partial^2}{\partial t^2} \int_0^r \iint_{S_\rho^M} u(x'_1, x'_2, x'_3, t) dS_\rho d\rho \\
&= \frac{\partial^2}{\partial t^2} \int_0^r \iint_{S_1^M} u(\rho, \theta, \varphi, t) d\omega \rho^2 d\rho \\
&= 4\pi \frac{\partial^2}{\partial t^2} \int_0^r \left(\frac{1}{4\pi} \iint_{S_1^M} u(\rho, \theta, \varphi, t) d\omega \right) \rho^2 d\rho \\
&= 4\pi \frac{\partial^2}{\partial t^2} \int_0^r \bar{u}(\rho, t) \rho^2 d\rho.
\end{aligned}$$

通解

左端=右端: 得到

$$4\pi \frac{\partial^2}{\partial t^2} \int_0^r \bar{u}(\rho, t) \rho^2 d\rho = 4\pi a^2 r^2 \frac{\partial \bar{u}(r, t)}{\partial r}.$$

两端对 r 求偏导数, 得到

$$4\pi \frac{\partial^2}{\partial t^2} (\bar{u}(r, t) r^2) = 4\pi a^2 \frac{\partial}{\partial r} (r^2 \frac{\partial \bar{u}(r, t)}{\partial r}),$$

所以

$$\frac{\partial^2 \bar{u}(r, t)}{\partial t^2} = \frac{a^2}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial \bar{u}(r, t)}{\partial r}) = \frac{1}{r} a^2 \frac{\partial^2 (r \bar{u}(r, t))}{\partial r^2}$$

写成

$$\frac{\partial^2 (r \bar{u})}{\partial t^2} = a^2 \frac{\partial^2 (r \bar{u})}{\partial r^2}$$

它的通解

$$r \bar{u}(r, t) = f_1(r + at) + f_2(r - at).$$

初始条件

由初始条件, 得

$$f_1(r) + f_2(r) = r\bar{u}(r, t)|_{t=0} = r\bar{\varphi}(r)$$

$$af_1'(r) - af_2'(r) = \frac{\partial}{\partial t}(r\bar{u}(r, t))|_{t=0} = r\bar{\psi}(r)$$

由此得

$$f_1(r) = \frac{1}{2} \left(r\bar{\varphi}(r) + \frac{1}{a} \int_0^r \rho\bar{\psi}(\rho) d\rho + C \right)$$

$$f_2(r) = \frac{1}{2} \left(r\bar{\varphi}(r) - \frac{1}{a} \int_0^r \rho\bar{\psi}(\rho) d\rho - C \right)$$

从而

$$\bar{u}(r, t) = \frac{(r+at)\bar{\varphi}(r+at) + (r-at)\bar{\varphi}(r-at)}{2r} + \frac{1}{2ar} \int_{r-at}^{r+at} \rho\bar{\psi}(\rho) d\rho$$

关于 $\mathbf{r}$ 作偶延拓

对 $\bar{u}(\mathbf{r}, t)$ 和初值 $\bar{\varphi}(\mathbf{r}, t), \bar{\psi}(\mathbf{r}, t)$ 作关于 $\mathbf{r}$ 作偶延拓

$$\bar{\varphi}(\mathbf{r}, t) \quad \text{和} \quad \bar{\psi}(\mathbf{r}, t).$$

事实上, 由于

$$\begin{aligned} \bar{u}(\mathbf{r}, t) &= \frac{1}{4\pi} \iint_{S_1^M} u(\xi, \eta, \zeta, t) d\omega \\ &= \frac{1}{4\pi} \iint_{\alpha_1^2 + \alpha_2^2 + \alpha_3^2 = 1} u(x_1 + r\alpha_1, x_2 + r\alpha_2, x_3 + r\alpha_3, t) d\omega \end{aligned}$$

有

$$\begin{aligned}
 \bar{u}(-r, t) &= \frac{1}{4\pi} \iint_{\alpha_1^2 + \alpha_2^2 + \alpha_3^2 = 1} u(x_1 - r\alpha_1, x_2 - r\alpha_2, x_3 - r\alpha_3, t) d\omega \\
 &= \frac{1}{4\pi} \iint_{\alpha_1^2 + \alpha_2^2 + \alpha_3^2 = 1} u(x_1 + r(-\alpha_1), x_2 + r(-\alpha_2), x_3 + r(-\alpha_3), t) d\omega \\
 &= \frac{1}{4\pi} \iint_{\beta_1^2 + \beta_2^2 + \beta_3^2 = 1} u(x_1 + r\beta_1, x_2 + r\beta_2, x_3 + r\beta_3, t) d\omega \\
 &= \bar{u}(r, t)
 \end{aligned}$$

所以 $\bar{u}(r, t)$ 是偶函数, r 的取值拓广到可以是实数。

令 $r \rightarrow 0$, 并利用 L'Hospital 法则, 有

$$\begin{aligned}
u(M, t) &= \bar{u}(0, t) = \lim_{r \rightarrow 0} \left[\frac{(at + r)\bar{\varphi}(at + r) - (at - r)\bar{\varphi}(at - r)}{2r} + \frac{1}{2ar} \int_{r-at}^{r+at} \rho \bar{\psi}(\rho) d\rho \right] \\
&= \lim_{r \rightarrow 0} \left[\frac{\bar{\varphi}(at + r) + (at + r)\bar{\varphi}'_0(at + r) + \bar{\varphi}(at - r) + (at - r)\bar{\varphi}'_0(at - r)}{2} \right. \\
&\quad \left. + \frac{(r + at)\bar{\psi}(r + at) - (r - at)\bar{\psi}(r - at)}{2a} \right] \\
&= \frac{2\bar{\varphi}(at) + 2at\bar{\varphi}'_0(at)}{2} + \frac{t\bar{\psi}(at) - t\bar{\psi}(-at)}{2} \\
&= \bar{\varphi}(at) + at\bar{\varphi}'_0(at) + t\bar{\psi}(at) \\
&= \frac{1}{a} \frac{\partial}{\partial t} [at\bar{\varphi}(at)] + t\bar{\psi}(at) \\
&= \frac{1}{4\pi a} \frac{\partial}{\partial t} \iint_{S_{at}^M} \frac{\varphi}{at} dS + \frac{1}{4\pi a} \iint_{S_{at}^M} \frac{\psi}{at} dS
\end{aligned}$$

Cauchy问题解

齐次三维波动方程Cauchy问题的 **Poisson 公式**

$$u(x, t) = \frac{1}{4\pi a} \frac{\partial}{\partial t} \left(\iint_{S_{at}(x)} \frac{\varphi}{at} dS \right) + \frac{1}{4\pi a} \iint_{S_{at}(x)} \frac{\psi}{at} dS$$

(3) 更一般地, 具有**非齐次项**的Cauchy问题解的**Kirchhoff公式**:

$$\begin{aligned} u(x, t) = & \frac{\partial}{\partial t} \left(\frac{1}{4\pi a^2 t} \int \int_{S_{at}(x)} \varphi(y) dS \right) + \frac{1}{4\pi a^2 t} \int \int_{S_{at}(x)} \psi(y) dS \\ & + \int_0^t \left(\frac{1}{4\pi a^2 (t - \tau)} \int \int_{S_{a(t-\tau)}(x)} f(y, \tau) dS \right) d\tau. \end{aligned}$$

其中: $S_t(x) = \{y \in \mathbb{R}^3 \mid |x - y| = t\}$.

同学们，再见！